

PIXELRAG FOR GEOGUESSR

Visual Retrieval for Image Geolocation and a Text-Aware Extension

Research Tryout Report

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Research question

How useful is PixelRAG-style visual retrieval for image geolocation, and which kinds of visual evidence make retrieval succeed or fail on GeoGuessr-style images?

Evaluation benchmark: 150 images • 30 countries • 5 images/country
Image-only retrieval • deterministic reader • OCR/text-aware extension

Abstract

This report evaluates PixelRAG-style visual retrieval as a source of geographic evidence for GeoGuessr-style image geolocation. A frozen benchmark of 150 images covering 30 countries was queried against the hosted PixelRAG service with Top-5 retrieval. Under the finalized Phase 6 comparison, 26/150 images (17.3%) contained correct country evidence in the Top-5. A deterministic rank-weighted reader made 26 correct predictions, 25 incorrect predictions, and abstained on 99 images, giving 17.3% overall accuracy and 34.0% coverage. A conservative OCR extension produced usable text for only 3/150 images (2.0%). Among those three, text-aware retrieval added correct country evidence in 1/3 cases (33.3%); the other two remained unresolved. The result supports text-aware retrieval as a complementary proof of concept, but does not establish an overall accuracy improvement.

1. Research Question and Motivation

The project investigates: “How useful is PixelRAG-style visual retrieval for image geolocation, and which kinds of visual evidence make retrieval succeed or fail on GeoGuessr-style images?” The study starts with a direct image-query baseline and then tests a text-aware extension in which visible text is extracted and used as a targeted retrieval query.

The goal is not to claim a complete GeoGuessr solver. It isolates whether visual retrieval can surface geographically informative evidence and whether a simple downstream reader can convert that evidence into country-level decisions.

2. Dataset and Evaluation Protocol

The source dataset was the Kaggle dataset “ubiquitin/geolocation-geoguessr-images-50k”. The project enumerated 49,997 unique image files across 124 countries and froze 150 images: 30 countries × 5 images, selected with random seed 42. Country-level evaluation was used because reliable city/region labels were not available in the frozen metadata.

Frozen countries: Andorra, Argentina, Botswana, Bulgaria, Cambodia, China, Ecuador, Estonia, Eswatini, Faroe Islands, France, Greenland, Latvia, Lebanon, Luxembourg, Montenegro, Northern Mariana Islands, Pakistan, Poland, Puerto Rico, Senegal, South Africa, South Korea, Switzerland, Taiwan, Tunisia, Uganda, United Arab Emirates, United Kingdom, United States.

The evaluation metadata contains `image_id`, `image_path`, `country`, `city_region`, `split`, and `useful_visible_text`. The `city_region` field was blank for the frozen sample and was not used as a quantitative target. The original `useful_visible_text` field was unannotated, so the later OCR experiment was evaluated independently.

3. System and Experimental Setup

PixelRAG 0.4.0 was installed in Python 3.13.2. The completed baseline used the hosted PixelRAG API rather than the approximately 217 GB local index. Each image was submitted as an image query and the Top-5 retrieved documents were stored as raw JSONL and flattened CSV. The baseline query did not use gold country labels.

4. Country-Evidence Metric

A retrieval is counted as correct only when a strict country alias for the gold country appears in the retrieved page title. This avoids false positives such as treating North Korea as evidence for South Korea. The metric is retrieval-to-evidence performance, not end-to-end geolocation accuracy.

An earlier Phase 4 artifact reports 27/150 images with any country evidence, while the finalized Phase 6 comparable evaluation reports 26/150. The report uses Phase 6 for the main result and explicitly preserves this discrepancy.

5. Image-Only Baseline Results

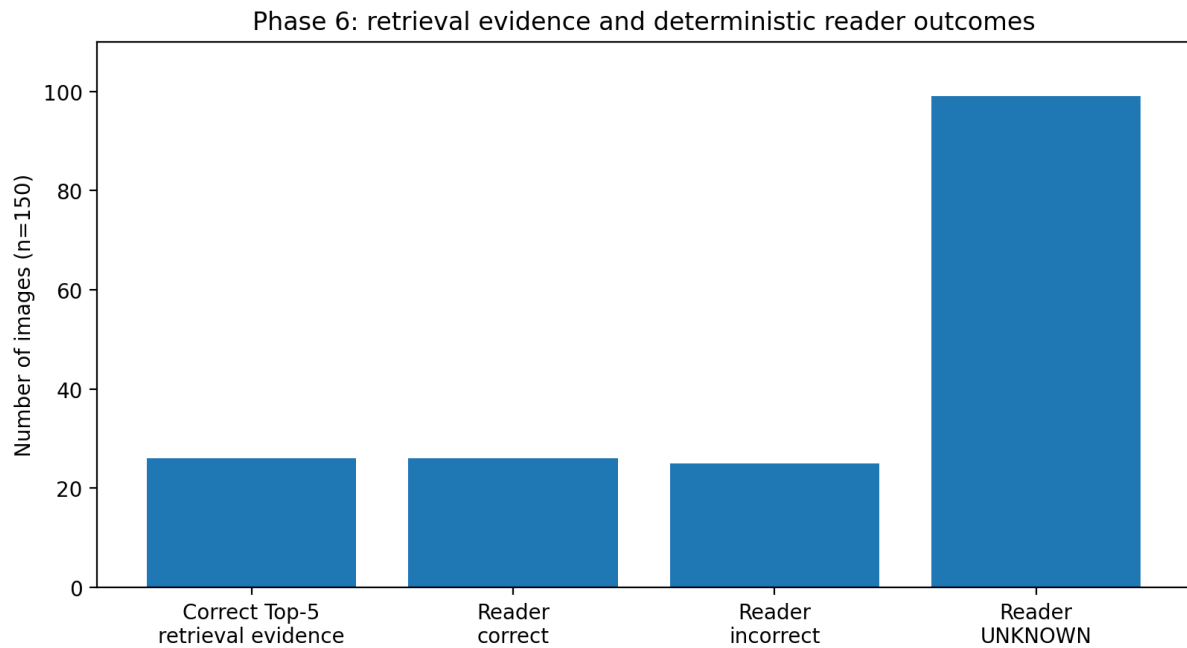


Figure 1. Phase 6 retrieval evidence and deterministic-reader outcomes.

Correct country evidence appeared in the Top-5 for 26/150 images (17.3%). The first correct evidence occurred at rank 1 for 7 images, rank 2 for 2, rank 3 for 5, rank 4 for 5, and rank 5 for 7.

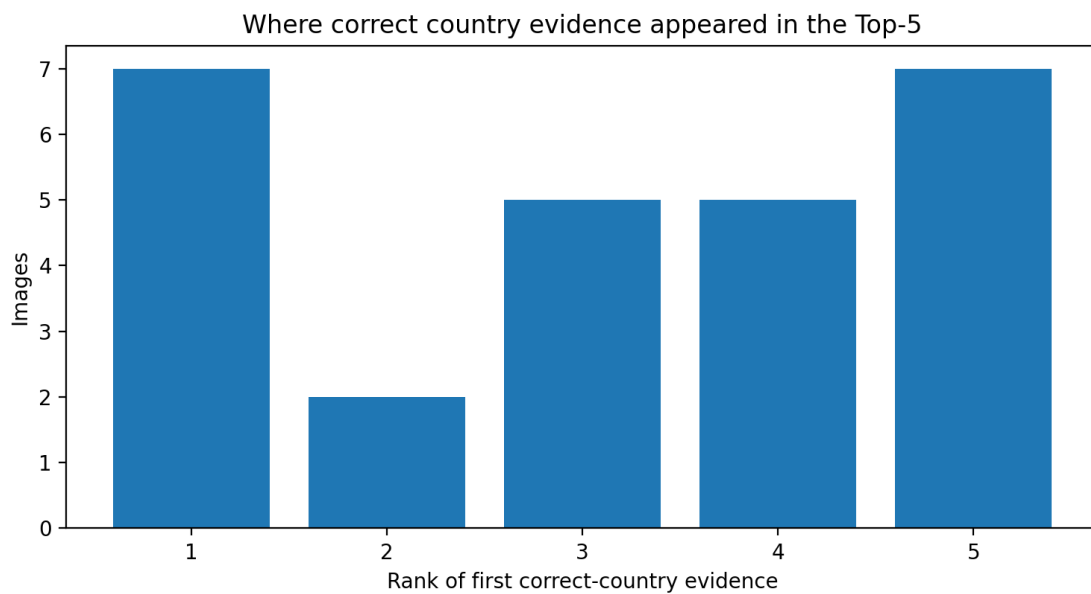


Figure 2. Distribution of the first correct-country evidence rank.

Measure	Result
Correct Top-5 evidence	26/150 (17.3%)
Reader correct	26/150 (17.3%)
Reader incorrect	25/150 (16.7%)
Reader UNKNOWN	99/150 (66.0%)
Reader coverage	34.0%
Selective accuracy	51.0%

The deterministic reader used rank weights 5, 4, 3, 2, 1 and returned UNKNOWN when no explicit country evidence was available or when there was a tie. All 26 images with correct Top-5 evidence became correct reader predictions. Among the 124 images without correct Top-5 evidence, the reader produced 25 incorrect predictions and 99 UNKNOWN results.

6. Failure Analysis

Phase 7 defines retrieval failure as no explicit gold-country evidence in the Top-5. This occurred for 124/150 images (82.7%). The automated categories were generated from retrieval titles and metadata only and were not treated as human-validated visual failure labels.

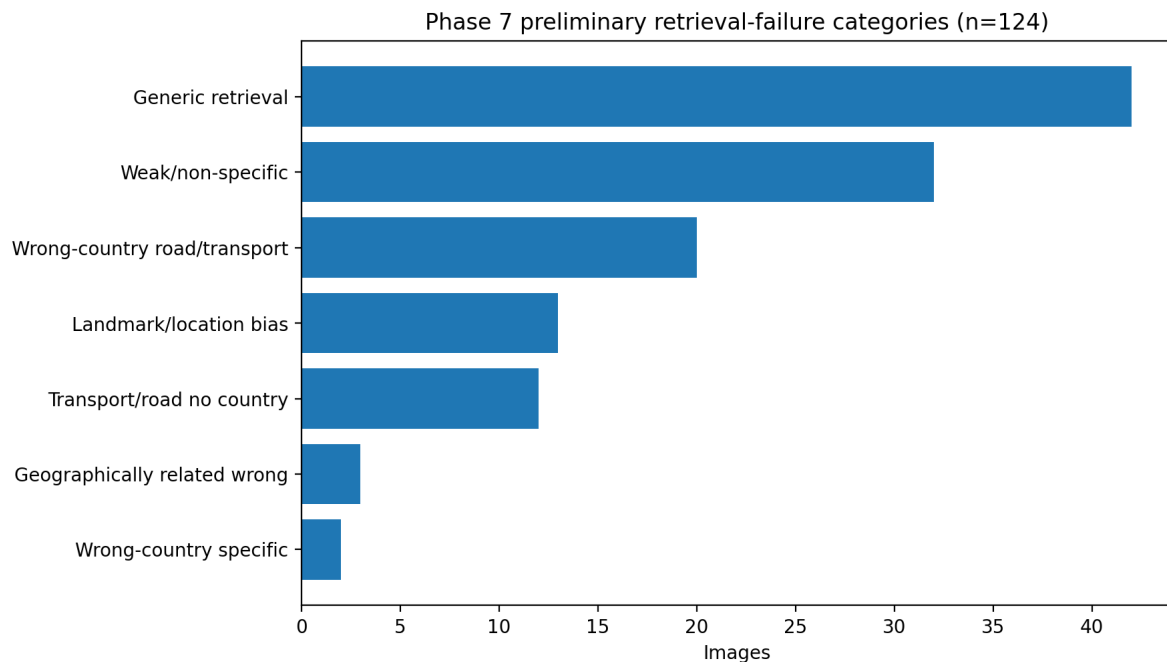


Figure 3. Preliminary automated retrieval-failure categories for the 124 no-correct-evidence cases.

The largest categories were generic retrieval (42; 33.9%), weak/non-specific retrieval (32; 25.8%), wrong-country transport/road evidence (20; 16.1%), and landmark/location bias (13; 10.5%).

The reader's 25 incorrect predictions were automatically categorized as 12 wrong transport/road clues, 5 generic clues, 5 wrong specific clues, and 3 geographically related clues. Forty-five retrieval cases were flagged for manual review. Visual ambiguity and missed text were not treated as automatically established causes.

7. Text-Aware OCR Extension

The extension used EasyOCR over a central gameplay crop, applied confidence/noise filters, removed common interface phrases, and issued a PixelRAG text query only when potentially useful text remained. All 150 images were processed.

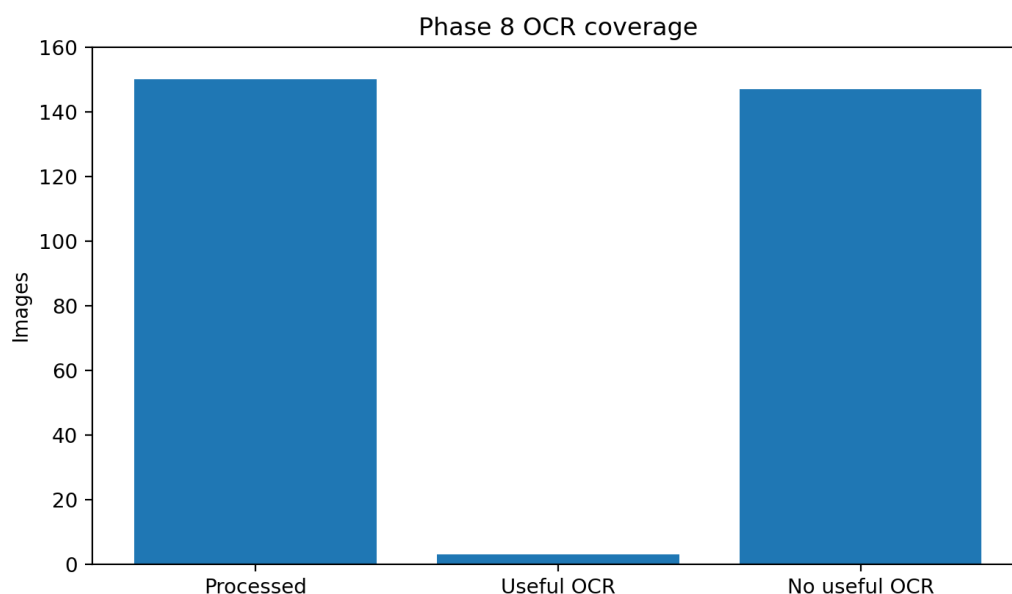


Figure 4. OCR coverage after conservative usefulness filtering.

Only 3/150 images (2.0%) produced text that passed the usefulness filters; 147/150 (98.0%) produced no useful OCR text. This low coverage is a major limitation.

8. Phase 9: Text-Aware versus Image-Only Retrieval

Across the three successful OCR cases, image-only retrieval had 0/3 correct evidence cases, while text-aware retrieval had 1/3. Thus the text-aware pipeline added correct country evidence in 1/3 cases (33.3%) within this small OCR-success subset.

France (geo_0055): OCR produced “liledeFrance moblites” and “Hybride”. Text-aware retrieval returned France-specific material including Plug-in_electric_vehicles_in_France at rank 3. Image-only retrieval had no correct France evidence.

South Korea (geo_0111): OCR produced “KOREA” and “coffee”. Retrieval included Coffee_in_South_Korea but also North_Korean_cuisine; strict matching therefore did not credit it as correct South Korea evidence.

Uganda (geo_0133): OCR produced “REAL TASK”. The text was not geographically informative and produced no correct Uganda evidence.

9. Discussion

PixelRAG can surface useful geographic evidence, but visual similarity frequently retrieves generic or geographically misleading documents. The reader results indicate that retrieval quality is the primary bottleneck of this simple pipeline: correct explicit evidence was converted correctly in all 26 observed cases, while absent correct evidence led only to abstention or error.

The OCR experiment shows that text can introduce evidence not surfaced by image retrieval, but the 2.0% useful-OCR coverage and three comparison cases prevent any claim of overall improvement.

10. Limitations

- The benchmark contains 150 images from 30 countries with five images per country.
- The primary metric measures explicit country evidence in page titles, not end-to-end geolocation accuracy.
- The deterministic reader is a simple rule-based reader, not a learned geolocation model.
- Phase 7 failure categories are automated preliminary labels and were not fully human-validated.
- City/region evaluation was not used because reliable labels were unavailable in the frozen metadata.
- Useful OCR was obtained for only 3/150 images.
- Phase 9 contains only three successful OCR cases; 1/3 should not be generalized.
- Ambiguous text such as “KOREA” was not credited as South Korea evidence under strict matching.
- The earlier Phase 4 27/150 artifact differs from the finalized Phase 6 26/150 comparison; this is explicitly documented.

11. Conclusion

On the frozen benchmark, correct country evidence appeared in the Top-5 for 26 images (17.3%), and the deterministic reader achieved 17.3% overall accuracy with 34.0% coverage. Correct retrieval evidence was sufficient for the reader in all observed cases, while absent correct evidence resulted in abstention or an incorrect prediction.

The text-aware extension provides an initial proof of concept: one of three successful OCR cases added correct country evidence where image-only retrieval had none. However, the 2.0% useful-OCR coverage and three-case comparison prevent a claim of overall improvement. The next research direction is to improve text detection, distinguish geographic text from UI/noise, and trigger targeted retrieval only when extracted clues are geographically specific.

12. Reproducibility

The project contains frozen metadata, retrieval records, flattened retrieval hits, reader predictions and metrics, failure-analysis outputs, OCR/text retrieval records, and Phase 9 comparison files. Source scripts are retained under src/ and configuration under configs/.

- python src\prepare_dataset.py --plan
- python src\prepare_dataset.py --download
- python src\query_pixelrag.py --limit 150
- python src\evaluate.py
- python src\run_reader.py
- python src\evaluate_reader.py
- python src\analyze_phase6.py

- `python src\analyze_phase7.py`
- `python src\text_aware.py --limit 150`
- `python src\evaluate_phase9.py`

The local PixelRAG index was intentionally not required; the completed retrieval baseline used the hosted PixelRAG service and retained raw retrieval outputs.

Appendix A. Frozen Evaluation Countries

Andorra, Argentina, Botswana, Bulgaria, Cambodia, China, Ecuador, Estonia, Eswatini, Faroe Islands, France, Greenland, Latvia, Lebanon, Luxembourg, Montenegro, Northern Mariana Islands, Pakistan, Poland, Puerto Rico, Senegal, South Africa, South Korea, Switzerland, Taiwan, Tunisia, Uganda, United Arab Emirates, United Kingdom, United States

Appendix B. Submission Artifact Inventory

- `data/eval_metadata.csv`
- `data/sample_plan.csv`
- `results/retrieval_records.jsonl`
- `results/retrieval_hits.csv`
- `results/phase6_metrics.json`
- `results/reader_predictions.csv`
- `results/phase7_failure_analysis.csv`
- `results/phase7_failure_summary.json`
- `results/text_aware_records.jsonl`
- `results/phase9_text_vs_image.csv`
- `results/phase9_metrics.json`